

Chebyshev Functions

1 Definitions and Basic Properties

Definition 1 (Prime Counting Function). *The prime counting function, denoted by $\pi(x)$, counts the number of primes less than or equal to the real number x :*

$$\pi(x) = \sum_{p \leq x} 1$$

Definition 2 (First Chebyshev Function). *The Chebyshev Theta function, or First Chebyshev function, denoted by $\theta(x)$, is defined as:*

$$\theta(x) = \sum_{p \leq x} \log p$$

where the sum is over all primes $p \leq x$.

Theorem 3. *For any real positive number x , one has:*

$$\theta(x) \leq x \log 4$$

In particular, $\theta(x) = O(x)$ as $x \rightarrow \infty$.

Proof. We first prove the inequality for any positive integer n by strong induction. Clearly, $\theta(1) = 0 < \log 4$ and $\theta(2) = \log 2 < 2 \log 4$, so the property is satisfied for $n = 1$ and $n = 2$. Assume the inequality holds for all integers $k < n$. We consider two cases for n :

Case 1: n is even. Let $n = 2m$ for some integer $m \geq 1$. Since $2m$ is not a prime for $m > 1$, we have $\theta(2m) = \theta(2m - 1)$. By our induction hypothesis, we get:

$$\theta(2m) = \theta(2m - 1) < (2m - 1) \log 4 \leq 2m \log 4$$

Case 2: n is odd. Let $n = 2m + 1$ for some integer $m \geq 1$. Consider the binomial coefficient $\binom{2m+1}{m}$. Since $\binom{2m+1}{m} = \binom{2m+1}{m+1}$, we have:

$$2^{2m+1} = \sum_{k=0}^{2m+1} \binom{2m+1}{k}$$

So,

$$2 \binom{2m+1}{m} \leq \sum_{k=0}^{2m+1} \binom{2m+1}{k} = 2^{2m+1} \implies \binom{2m+1}{m} \leq 2^{2m} = 4^m$$

On the other hand, we can write the binomial coefficient as:

$$\binom{2m+1}{m} = \frac{(2m+1)!}{m!(m+1)!} = \frac{(2m+1)(2m) \dots (m+2)}{1 \cdot 2 \dots m}$$

Any prime p such that $m+1 < p \leq 2m+1$ divides the numerator $(2m+1)!$ but is strictly greater than any factor in the denominator $m!(m+1)!$. Therefore:

$$\prod_{m+1 < p \leq 2m+1} p \leq \binom{2m+1}{m} \leq 4^m$$

Taking the logarithm of both sides yields:

$$\theta(2m+1) - \theta(m+1) = \sum_{m+1 < p \leq 2m+1} \log p \leq m \log 4$$

Hence, using the induction hypothesis for $m + 1 < 2m + 1$:

$$\theta(2m + 1) \leq m \log 4 + \theta(m + 1) \leq m \log 4 + (m + 1) \log 4 = (2m + 1) \log 4$$

This completes the induction. For any real number $x > 0$, we have $\theta(x) = \theta(\lfloor x \rfloor)$. Thus:

$$\theta(x) = \theta(\lfloor x \rfloor) \leq \lfloor x \rfloor \log 4 \leq x \log 4$$

Combining the fact that $\theta(x) \geq 0$ with $\theta(x) \leq x \log 4$, we get $\theta(x) = O(x)$. $\square$

Definition 4 (Second Chebyshev Function). *The Chebyshev Psi function, denoted by $\psi(x)$, is defined as:*

$$\psi(x) = \sum_{n \leq x} \Lambda(n)$$

where $\Lambda(n)$ is the von Mangoldt function.

Remark 5. We can rewrite $\psi(x)$ as:

$$\psi(x) = \sum_{p^k \leq x} \log p = \sum_{p \leq x} \left\lfloor \frac{\log x}{\log p} \right\rfloor \log p$$

2 Relations Connecting $\theta(x)$, $\psi(x)$, and $\pi(x)$

Theorem 6. *The following relations hold:*

$$\pi(x) = \frac{\theta(x)}{\log x} + \int_2^x \frac{\theta(t)}{t \log^2 t} dt = \frac{\theta(x)}{\log x} + O\left(\frac{x}{\log^2 x}\right)$$

and

$$\theta(x) = \pi(x) \log x - \int_2^x \frac{\pi(t)}{t} dt = \pi(x) \log x + O\left(\frac{x}{\log x}\right)$$

Proof. First, we apply Abel's partial summation formula. Let $a_n = \log n$ if n is prime, and 0 otherwise. Then $A(x) = \sum_{n \leq x} a_n = \theta(x)$. Let $f(t) = \frac{1}{\log t}$. Then:

$$\begin{aligned} \pi(x) &= \sum_{p \leq x} 1 = \sum_{n \leq x} a_n f(n) = A(x) f(x) - \int_2^x A(t) f'(t) dt \\ \pi(x) &= \frac{\theta(x)}{\log x} - \int_2^x \theta(t) \left(-\frac{1}{t \log^2 t} \right) dt = \frac{\theta(x)}{\log x} + \int_2^x \frac{\theta(t)}{t \log^2 t} dt \end{aligned}$$

Since $\theta(x) = O(x)$, the integral is bounded by $\int_2^x \frac{O(t)}{t \log^2 t} dt = O\left(\frac{x}{\log^2 x}\right)$. The formula for $\theta(x)$ is obtained similarly by setting $a_n = 1$ if n is prime and 0 otherwise, giving $A(x) = \pi(x)$, and taking $f(t) = \log t$. $\square$

Corollary 7. *The asymptotic relation $\pi(x) \sim \frac{x}{\log x}$ as $x \rightarrow \infty$ is equivalent to $\theta(x) \sim x$ as $x \rightarrow \infty$.*

Theorem 8. *The following relations hold:*

$$\begin{aligned} \psi(x) &= \sum_{k \geq 1} \theta(x^{\frac{1}{k}}) = \theta(x) + O(\sqrt{x} \log x) \\ \theta(x) &= \sum_{k \geq 1} \mu(k) \psi(x^{\frac{1}{k}}) \end{aligned}$$

Proof. By definition,

$$\psi(x) = \sum_{p^k \leq x} \log p = \sum_{k \geq 1} \sum_{p \leq x^{\frac{1}{k}}} \log p = \sum_{k \geq 1} \theta(x^{\frac{1}{k}})$$

This sum is finite. Since $x^{\frac{1}{k}} < 2$ and $\theta(x^{\frac{1}{k}}) = 0$ when $k > \frac{\log x}{\log 2}$, we have:

$$\psi(x) = \sum_{k=1}^{\lfloor \frac{\log x}{\log 2} \rfloor} \theta(x^{\frac{1}{k}}) = \theta(x) + \sum_{k=2}^{\lfloor \frac{\log x}{\log 2} \rfloor} \theta(x^{\frac{1}{k}})$$

For $k \geq 2$, $\theta(x^{\frac{1}{k}}) = O(x^{\frac{1}{k}})$. The number of terms in the last sum is bounded by $\lfloor \frac{\log x}{\log 2} \rfloor = O(\log x)$. Furthermore, $x^{\frac{1}{k}}$ is strictly decreasing with respect to k , so the largest term occurs when $k = 2$. Thus:

$$\psi(x) - \theta(x) \leq \left\lfloor \frac{\log x}{\log 2} \right\rfloor \theta(\sqrt{x}) = O(\sqrt{x} \log x)$$

For the second relation, we apply Möbius inversion. We have:

$$\psi(x^{\frac{1}{k}}) = \sum_{m \geq 1} \theta\left((x^{\frac{1}{k}})^{\frac{1}{m}}\right) = \sum_{m \geq 1} \theta(x^{\frac{1}{km}})$$

Hence,

$$\sum_{k \geq 1} \mu(k) \psi(x^{\frac{1}{k}}) = \sum_{k \geq 1} \mu(k) \sum_{m \geq 1} \theta(x^{\frac{1}{km}})$$

Let $n = km$. By rearranging the summation, we get:

$$\sum_{n \geq 1} \theta(x^{\frac{1}{n}}) \sum_{k|n} \mu(k)$$

Since $\sum_{k|n} \mu(k)$ is 1 if $n = 1$ and 0 otherwise, the sum collapses entirely to $\theta(x)$. $\square$

Corollary 9. *The asymptotic relation $\theta(x) \sim x$ as $x \rightarrow \infty$ is equivalent to $\psi(x) \sim x$ as $x \rightarrow \infty$.*

3 Legendre's Formula and Bertrand's Postulate

Definition 10. *Let n be a positive integer and p a prime number. We denote the p -adic valuation of n by $v_p(n) = j$, where $j \in \mathbb{N}$ if $p^j \mid n$ and $p^{j+1} \nmid n$. Hence, $\forall n \in \mathbb{N}$, $n = \prod_p p^{v_p(n)}$.*

Theorem 11 (Legendre's Formula). *Let n be a positive integer and let p be a prime number. Then:*

$$v_p(n!) = \sum_{k \geq 1} \left\lfloor \frac{n}{p^k} \right\rfloor$$

Proof. Note that this sum is finite since if $p^k > n$ (i.e., $k > \frac{\log n}{\log p}$), then $\lfloor \frac{n}{p^k} \rfloor = 0$. By definition:

$$v_p(n!) = v_p\left(\prod_{m=1}^n m\right) = \sum_{m=1}^n v_p(m)$$

To calculate $\sum v_p(m)$ for $m = 1, \dots, n$, we can count the number of multiples of $p, p^2, \dots$ up to n . For any positive integer k , the sequence $1, 2, \dots, n$ contains exactly $\lfloor \frac{n}{p^k} \rfloor$ multiples of p^k . The multiples are $p^k, 2p^k, \dots, \lfloor \frac{n}{p^k} \rfloor p^k$.

Consider a specific integer m in the range $1 \dots n$. Suppose that the exact highest power of p dividing m is p^j for some $j \geq 0$, meaning $v_p(m) = j$. We must verify that this integer m contributes exactly j to the sum $\sum_{k \geq 1} \lfloor \frac{n}{p^k} \rfloor$. Because m is a multiple of p , it is counted once in $\lfloor \frac{n}{p} \rfloor$. It is a multiple of p^2 , so it is counted once in $\lfloor \frac{n}{p^2} \rfloor$, and so on, up to being counted once in $\lfloor \frac{n}{p^j} \rfloor$. However, because m is not a multiple of p^{j+1} or any higher power, it is not counted for any $k > j$. Thus, summing over all integers m from 1 to n , we obtain exactly the proposed formula. $\square$

Theorem 12 (Bertrand's Postulate). *For every positive integer $n \geq 1$, there is at least one prime p such that $n < p \leq 2n$.*

Proof. (Due to Erdős). Assume for contradiction that for some integer $n \geq 1$, there is no prime p such that $n < p \leq 2n$. We consider the central binomial coefficient $\binom{2n}{n} = \frac{(2n)!}{(n!)^2}$. By Legendre's formula, the prime factorization of $\binom{2n}{n}$ is $\prod_{p \leq 2n} p^{R(p,n)}$, where:

$$R(p, n) = v_p((2n)!) - 2v_p(n!) = \sum_{k \geq 1} \left(\left\lfloor \frac{2n}{p^k} \right\rfloor - 2 \left\lfloor \frac{n}{p^k} \right\rfloor \right)$$

Notice that each term in the sum is either 0 or 1. Moreover, the sum vanishes for $k > \lfloor \log_p(2n) \rfloor$. Thus, $R(p, n) \leq \lfloor \log_p(2n) \rfloor$, which implies $p^{R(p,n)} \leq 2n$.

We can bound the prime contributions as follows:

- For $p > \sqrt{2n}$, we have $\lfloor \log_p(2n) \rfloor = 1$, so $R(p, n) \leq 1$.
- If $\frac{2n}{3} < p \leq n$, then $2p \leq 2n < 3p$, meaning $\lfloor \frac{2n}{p} \rfloor = 2$ and $\lfloor \frac{n}{p} \rfloor = 1$. Thus $R(p, n) = 2 - 2(1) = 0$.
- By our assumption, there are no primes $n < p \leq 2n$.

Consequently, $\binom{2n}{n}$ has no prime factors greater than $\frac{2n}{3}$. Its prime factorization separates into primes $p \leq \sqrt{2n}$ and primes $\sqrt{2n} < p \leq \frac{2n}{3}$. Using the bound $p^{R(p,n)} \leq 2n$ for the former, and $R(p, n) \leq 1$ for the latter, we obtain:

$$\binom{2n}{n} \leq \prod_{p \leq \sqrt{2n}} 2n \prod_{\sqrt{2n} < p \leq \frac{2n}{3}} p$$

Since there are at most $\sqrt{2n}$ primes up to $\sqrt{2n}$, the first product is bounded by $(2n)^{\sqrt{2n}}$. For the second product, recall that $\theta(x) = \sum_{p \leq x} \log p \leq x \log 4$, which means $\prod_{p \leq x} p \leq 4^x$. Thus, the second product is bounded by $4^{2n/3}$. Combining these, we have:

$$\binom{2n}{n} \leq (2n)^{\sqrt{2n}} 4^{2n/3}$$

On the other hand, $4^n = (1+1)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k}$. The largest term in this sum of $2n+1$ elements is $\binom{2n}{n}$, so:

$$\frac{4^n}{2n+1} \leq \binom{2n}{n}$$

Putting the inequalities together yields:

$$\frac{4^n}{2n+1} \leq (2n)^{\sqrt{2n}} 4^{2n/3} \implies 4^{n/3} \leq (2n+1)(2n)^{\sqrt{2n}} \leq (2n)^{\sqrt{2n}+1}$$

Taking the logarithm, we get $\frac{n}{3} \log 4 \leq (\sqrt{2n} + 1) \log(2n)$. This inequality fails for sufficiently large n (specifically, for $n \geq 4000$). For $n < 4000$, we can explicitly find a sequence of primes where each is less than twice the previous: 2, 3, 5, 7, 13, 23, 43, 83, 163, 317, 631, 1259, 2503, 4001. This sequence guarantees a prime in $(n, 2n]$ for all $n < 4000$. Thus, Bertrand's postulate holds for all $n \geq 1$. $\square$

4 Mertens' Theorems

Theorem 13 (Mertens' First Theorem). *For any real number $x \geq 2$, we have:*

$$\sum_{\substack{p \leq x \\ p \text{ prime}}} \frac{\log p}{p} = \log x + O(1)$$

Note: all mentions of p in the upcoming proof is for **prime numbers**.

Proof. By Legendre's formula, $v_p(n!) = \sum_{k \geq 1} \lfloor \frac{n}{p^k} \rfloor$. Taking the logarithm of $n!$:

$$\begin{aligned} \log(n!) &= \log \left(\prod_{p \leq n} p^{v_p(n!)} \right) = \sum_{p \leq n} v_p(n!) \log p \\ &= \sum_{p \leq n} \log p \sum_{k \geq 1} \left\lfloor \frac{n}{p^k} \right\rfloor = \sum_{p \leq n} \log p \left\lfloor \frac{n}{p} \right\rfloor + \sum_{p \leq n} \sum_{k \geq 2} \left\lfloor \frac{n}{p^k} \right\rfloor \log p \end{aligned}$$

We bound the second sum. Since $\lfloor y \rfloor \leq y$, we have:

$$\sum_{p \leq n} \sum_{k \geq 2} \left\lfloor \frac{n}{p^k} \right\rfloor \log p \leq n \sum_{p \leq n} \log p \sum_{k \geq 2} \frac{1}{p^k} = n \sum_{p \leq n} \frac{\log p}{p(p-1)}$$

Since the series $\sum_{m=2}^{\infty} \frac{\log m}{m(m-1)}$ is absolutely convergent, the entire series is bounded by $O(n)$. Thus:

$$\log(n!) = \sum_{p \leq n} \left\lfloor \frac{n}{p} \right\rfloor \log p + O(n) = \sum_{p \leq n} \left(\frac{n}{p} + O(1) \right) \log p + O(n)$$

$$= n \sum_{p \leq n} \frac{\log p}{p} + O\left(\sum_{p \leq n} \log p\right) + O(n) = n \sum_{p \leq n} \frac{\log p}{p} + O(\theta(n)) + O(n)$$

Since $\theta(n) = O(n)$, this becomes:

$$\log(n!) = n \sum_{p \leq n} \frac{\log p}{p} + O(n)$$

On the other hand, by Stirling's formula, $\log(n!) = n \log n - n + O(\log n) = n \log n + O(n)$. Hence:

$$n \sum_{p \leq n} \frac{\log p}{p} + O(n) = n \log n + O(n)$$

Dividing by n , we conclude:

$$\sum_{p \leq n} \frac{\log p}{p} = \log n + O(1)$$

□

Theorem 14 (Mertens' Second Theorem). *There exists a constant M (known as the Meissel-Mertens constant) such that:*

$$\sum_{p \leq x} \frac{1}{p} = \log \log x + M + O\left(\frac{1}{\log x}\right)$$

Note: all mentions of p in the upcoming proof is for **prime numbers**.

Proof. Let $A(x) = \sum_{p \leq x} \frac{\log p}{p}$. From Mertens' First Theorem, we know that $A(x) = \log x + R(x)$ where $R(x) = O(1)$. Using partial summation with $a_n = \frac{\log n}{n}$ if n is prime and 0 otherwise, and $f(t) = \frac{1}{\log t}$, we get:

$$\begin{aligned} \sum_{p \leq x} \frac{1}{p} &= \frac{A(x)}{\log x} + \int_2^x \frac{A(t)}{t \log^2 t} dt \\ &= \frac{\log x + R(x)}{\log x} + \int_2^x \frac{\log t + R(t)}{t \log^2 t} dt \\ &= 1 + O\left(\frac{1}{\log x}\right) + \int_2^x \frac{1}{t \log t} dt + \int_2^x \frac{R(t)}{t \log^2 t} dt \end{aligned}$$

Evaluating the first integral:

$$\int_2^x \frac{1}{t \log t} dt = \log \log x - \log \log 2$$

For the second integral, let $C_0 = \int_2^\infty \frac{R(t)}{t \log^2 t} dt$, which converges since $R(t) = O(1)$. We can write:

$$\int_2^x \frac{R(t)}{t \log^2 t} dt = C_0 - \int_x^\infty \frac{R(t)}{t \log^2 t} dt = C_0 - O\left(\int_x^\infty \frac{1}{t \log^2 t} dt\right) = C_0 - O\left(\frac{1}{\log x}\right)$$

Finally, combining these terms gives:

$$\sum_{p \leq x} \frac{1}{p} = \log \log x + (1 - \log \log 2 + C_0) + O\left(\frac{1}{\log x}\right)$$

Setting $M = 1 - \log \log 2 + C_0$ completes the proof. □

Theorem 15 (Mertens' Third Theorem). *As $x \rightarrow \infty$,*

$$\prod_{p \leq x} \left(1 - \frac{1}{p}\right) = \frac{e^{-\gamma}}{\log x} \left(1 + O\left(\frac{1}{\log x}\right)\right)$$

Note: all mentions of p in the upcoming proof is for **prime numbers**.

Proof. Taking the negative natural logarithm of the product gives:

$$-\sum_{p \leq x} \log \left(1 - \frac{1}{p}\right) = \sum_{p \leq x} \sum_{k=1}^{\infty} \frac{1}{kp^k} = \sum_{p \leq x} \frac{1}{p} + \sum_{p \leq x} \sum_{k=2}^{\infty} \frac{1}{kp^k}$$

Let $C_1 = \sum_p \sum_{k=2}^{\infty} \frac{1}{kp^k}$. This series converges absolutely. The tail of the series can be bounded as:

$$\sum_{p > x} \sum_{k=2}^{\infty} \frac{1}{kp^k} \leq \sum_{p > x} \frac{1}{p(p-1)} = O\left(\frac{1}{x}\right)$$

Thus, the sum becomes:

$$\sum_{p \leq x} \frac{1}{p} + C_1 - O\left(\frac{1}{x}\right)$$

Applying Mertens' Second Theorem yields:

$$-\sum_{p \leq x} \log \left(1 - \frac{1}{p}\right) = \log \log x + M + C_1 + O\left(\frac{1}{\log x}\right)$$

Exponentiating both sides and multiplying by -1 , we obtain:

$$\begin{aligned} \prod_{p \leq x} \left(1 - \frac{1}{p}\right) &= \exp \left(-\log \log x - (M + C_1) + O\left(\frac{1}{\log x}\right) \right) \\ &= \frac{e^{-(M+C_1)}}{\log x} \exp \left(O\left(\frac{1}{\log x}\right) \right) = \frac{e^{-(M+C_1)}}{\log x} \left(1 + O\left(\frac{1}{\log x}\right) \right) \end{aligned}$$

A well-known consequence of the analytic properties of the Riemann zeta function links these constants, specifically establishing that $M + C_1 = \gamma$, where γ is the Euler-Mascheroni constant. Therefore, the constant term $e^{-(M+C_1)}$ simplifies to $e^{-\gamma}$, completing the proof. $\square$

Corollary 16. $M = \lim_{x \rightarrow \infty} \left(\sum_{p \leq x} \frac{1}{p} - \log \log x \right)$

Remark 17. $M = \gamma + \sum_{k=2}^{\infty} \frac{\mu(k)}{k} \log \zeta(k)$